

Syed Faizan Hussain Shah

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ABOUT ME

Master's student in Artificial Intelligence at Jeju National University, with a Bachelor's degree in Computer Science from COMSATS University Islamabad. My research focuses on Large Language Models (LLMs) and Vision-Language Models (VLMs), with emphasis on quantization, model compression, lightweight fine-tuning, efficient inference, and multilingual evaluation. I have experience designing and evaluating AI methods through controlled experiments and benchmarking, including research on low-resource multilingual VLMs and AI-driven railway infrastructure monitoring. My technical expertise includes Python, machine learning, deep learning, experimental design, model evaluation, and scientific writing. My research interests include efficient AI, multimodal and multilingual learning, and practical AI systems for resource-constrained environments.

RESEARCH INTEREST

Efficient AI & Model Compression · LLM/VLM Quantization · Multimodal & Multilingual AI · Lightweight Fine-Tuning · Efficient Inference · Smart Infrastructure & Transportation · Computer Vision · NLP

EDUCATION AND TRAINING

Master of Artificial Intelligence

Jeju National University [01/09/2025 – Current]

City: Jeju Special Self-Governing Province | **Country:** South Korea | **Website:** https://jejunu.ac.kr/

- Advanced research in **LLMs/VLMs** with a focus on quantization techniques
- Post-training quantization methods and quantization-aware training
- Efficient model compression, optimization and fine-tuning and adaptation of transformer-based architecture
- Scientific writing, paper review, and research methodology

BS Computer science

Comsats university Islamabad Attock campus [06/02/2019 – 31/01/2023]

City: Attock | **Country:** Pakistan | **Website:** http://attock.comsats.edu.pk

Fsc (Pre-engineering)

FBISE [15/08/2016 – 18/12/2018]

City: Islamabad | **Country:** Pakistan | **Website:** https://www.fbise.edu.pk

SSC

Government High School Hasan Abdal [09/04/2014 – 20/07/2016]

City: Attock | **Country:** Pakistan

WORK EXPERIENCE

– Changsha, China

Research Assistant

[10/2023 – 08/2025]

- Conducted research on AI-driven railway infrastructure monitoring and maintenance systems
- Developed and implemented deep learning and computer vision models for real-world applications
- Worked on synthetic data generation and point-cloud data processing for infrastructure analysis
- Performed data analysis, visualization, and experimental evaluation of machine learning models
- Collaborated on interdisciplinary research projects and contributed to scientific publications

- Utilized Python and modern deep learning frameworks for model development and optimization
- Demonstrated strong research, analytical, and problem-solving skills in applied AI projects

– Jeju Special Self-Governing Province, South Korea

Teaching Assistant

[03/2026 – Current]

Assisted in teaching undergraduate LLM-based coursework in **Generative AI** and **Big Data Analysis and Visualization**
Guided students on topics:

- Large Language Models (LLMs), prompt engineering, quantization and fine-tuning
- Supported in multimodal models (e.g., CLIP) or NLP systems

– Remote | Freelance

Academic Infographic Designer

[01/2025 – Current]

JOURNAL PUBLICATIONS

Sensitivity-aligned Mixed-precision Quantization and Low-rank Adaptation for LLMs

Under Review in Springer Nature Journal of Scientific Reports as first author.

Empirical Study of Constraint Effects in PlanRAG-Style LLM Agents for Decision QA

Under Review in International Journal on Advanced Science, Engineering and Information Technology as first author.

Arabic Sentiment Analysis and Sarcasm Detection using Probabilistic Projections Based Variational Switch Transformer

Published in Journal of IEEE Access as co-author

Link: https://ieeexplore.ieee.org/document/10163795

LiDAR-Simulated Contrastive 3D Construction Approach for Probabilistic Point Clouds Modeling of Rail Fasteners

Published in Journal of Computing in Civil Engineering as co-author.

Link: https://doi.org/10.1061/JCCEE5.CPENG-6137

Efficient and Interpretable Image Processing Approach for Crack Segmentation

Published in Journal of Infrastructure Systems as co-author.

Link: https://doi.org/10.1061/JITSE4.ISENG-2759

Self-supervised contrastive anomaly detection in railway fasteners using point clouds and deep metric learning for imbalance dataset

Published in Journal of Civil Structural Health Monitoring as co-author.

Link: https://doi.org/10.1007/s13349-025-00960-8

DA-NGF: A Domain-Adaptive Neurogenerative Framework for Few-Shot Railway Fastener Defect Synthesis and Transferable Representation Learning

Published in Journal of Applied Intelligence as co-author.

Link: https://doi.org/10.1007/s10489-025-07054-4

Multimodal Geometric AutoEncoder (MGAE) for Rail Fasteners Tightness Evaluation with Point Clouds & Monocular Depth Fusion

Published in Journal of Measurement as co-author.

Link: https://doi.org/10.1016/j.measurement.2024.116557

TrinityNet: A Novel Cross-Modality Generative Reasoning Framework for Zero-Label Railway Fastener Tightness Evaluation

Published in Journal of IET Intelligent Transport Systems as co-author.

Link: https://doi.org/10.1049/itr2.70164

Transforming railway transportation: the role of emerging technologies in efficiency, safety, and sustainability

Published in Journal of Civil Engineering and Management as co-author.

Link: https://doi.org/10.3846/jcem.2026.25811

CONFERENCE PUBLICATIONS

Romanization Matters in Multilingual Vision–Language Models

Submitted in **ARR October 2026** cycle for Conference (**NAACL**).

Preliminary Study on Mixed-Bit Quantization and Adaptive qLoRA⁺ Fine-Tuning for Efficient LLM

Published and presented in 12th International Conference on Internet of Things and Convergence 2026 as first author.

Also **Awarded as Best Research Paper** of the conference.

Quantization-Aware LLM Deployment for Resource-Constrained Onboard Aerospace Systems

Published and presented in 3rd IAA Conference on AI for Space 2026 as first author.

A Transparent CAPTCHAS Verification System for Cloud-based Smart & Secure Applications

Published and presented in 2023 IEEE International Smart Cities Conference (isc2) as first author.

Link: https://ieeexplore.ieee.org/document/10293665

Traffic Forecasting & Route Optimization in Smart Environment Using Graph Representation Learning

Published in 2023 IEEE International Smart Cities Conference (isc2).

Link: https://ieeexplore.ieee.org/document/10293287

PROJECTS

Undergraduate Final Year Project (Smart AI Diet Consultant) A diet suggesting android application was my undergraduate final year project which work on human details such as weight, height, age, gender, diabetes (Yes/No) and Goal (healthy/weight gain/weight loss) of user by that data Machine Learning model predict diet for user especially against diabetes.

Link: https://github.com/Fazi743/Smart-Dietition_FYP

Bus Ticket Booking System Bus Ticket Management System as part of my web development coursework in bachlor degree. Using HTML, CSS, and PHP.

Link: https://github.com/Fazi743/Bus-Ticket-booking-System

COVID Vaccine Sentiment Analysis This project focuses on analyzing sentiments expressed in tweets related to COVID-19 vaccines. By leveraging natural language processing (NLP) techniques, we aim to gain insights into public opinions, concerns, and attitudes toward COVID-19 vaccines.

Link: https://github.com/Fazi743/COVID-Vaccine-Sentiment-Analysis

SKILLS

Programming

Python / React Native / SQL

AI

Deep Learning / Machine Learning / Quantization / Natural Language Processing / Multimodal Models / Fine-Tuning

Frameworks & Libraries

TensorFlow / PyTorch, Keras / Pandas Numpy Matplotlib / Hugging Face Transformers

Research Skills

Experimental Design / Benchmarking / Scientific Writing / LaTeX/Overleaf / Statistical Analysis / Draw.io / MS Visio

LANGUAGE SKILLS

Mother tongue(s): Punjabi

Other language(s):

Urdu

LISTENING C2 **READING** C2 **WRITING** C2

SPOKEN PRODUCTION C2 **SPOKEN INTERACTION** C2

English

LISTENING C2 **READING** C2 **WRITING** C2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

Mandarin

LISTENING A1 **SPOKEN INTERACTION** A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

CERTIFICATION'S

DataCamp Course's

- Introduction to SQL
- Intermediate to SQL
- Joining Data in SQL
- Introduction to Python
- Introduction to Data Science

Coursera Course's

- AI For Everyone
- Programming With Python
- Python Data Structure
- Python For Data Science & AI Development
- Python Project For Data Science
- Data Analytics Essential
- Data Visualization & Dashboard With Excel
- Using Python to Access Web Data
- Using Databases with Python

IBM Course's

- Introduction to Data Analytics
- Excel Essential for Data Analytics

HOBBIES AND INTERESTS

Playing Cricket & Badminton | Watching Sports | Traveling

REFERENCES

Available upon request.